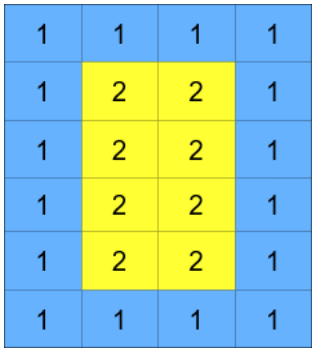
1914. Cyclically Rotating a Grid

Solved

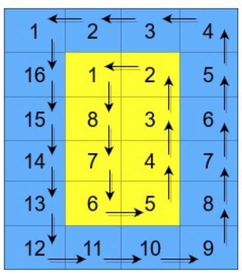
Medium Topics Hint

You are given an `m x n` integer matrix `grid`, where `m` and `n` are both **even** integers, and an integer `k`.

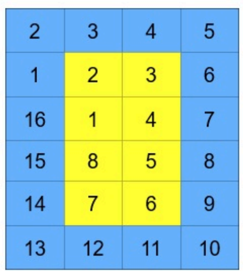
The matrix is composed of several layers, which is shown in the below image, where each color is its own layer:



A cyclic rotation of the matrix is done by cyclically rotating **each layer** in the matrix. To cyclically rotate a layer once, each element in the layer will take the place of the adjacent element in the **counter-clockwise** direction. An example rotation is shown below:



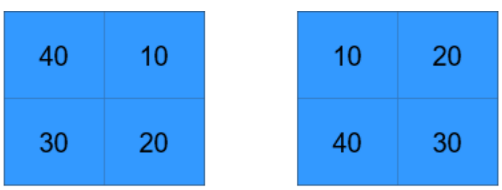
Before rotation



After rotation with k = **1**

Return *the matrix after applying* `k` *cyclic rotations to it*.

Example 1:



Before Any Rotations **After One Rotation**

> **Input:** grid = [[40,10],[30,20]], k = 1
> **Output:** [[10,20],[40,30]]
> **Explanation:** The figures above represent the grid at every state.

Example 2:

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Before Any Rotations

2	3	4	8
1	7	11	12
5	6	10	16
9	13	14	15

After One Rotation

3	4	8	12
2	11	10	16
1	7	6	15
5	9	13	14

After Two Rotations

> **Input:** grid = [[1,2,3,4],[5,6,7,8],[9,10,11,12],[13,14,15,16]], k = 2
> **Output:** [[3,4,8,12],[2,11,10,16],[1,7,6,15],[5,9,13,14]]
> **Explanation:** The figures above represent the grid at every state.

Constraints:

- `m == grid.length`
- `n == grid[i].length`
- `2 <= m, n <= 50`
- Both `m` and `n` are **even** integers.
- `1 <= grid[i][j] <= 5000`
- `1 <= k <= 10`^{9}

Seen this question in a real interview before? 1/6

Yes No

Accepted 79,216 /106.8K | Acceptance Rate 74.2 %

Topics

Hint 1

Hint 2

Discussion (96)